

PRELIMINARY ECOLOGICAL APPRAISAL

Preliminary Ecological Appraisal - date test project

Prepared For	Client
Site Reference	DTP-2026-763
Report Version	Version 1
Date	11/3/2026

TABLE OF CONTENTS

1.	Introduction	3
2.	Methodology	4
3.	Results	5
4.	Ecological Constraints	6
5.	Discussion & Conclusions	7
6.	Appendices	8

APPENDICES

Appendix A	Habitat Map
Appendix B	Species List
Appendix C	Site Photographs

1. Introduction

The purpose of this Preliminary Ecological Appraisal (PEA) is to assess the ecological value and potential constraints associated with the proposed development at Glenbeigh, County Kerry. The project, designated as the "date test project" with the site code DTP-2026-763, is situated at grid reference A 535 750. This appraisal aims to inform the planning process by identifying any ecological sensitivities and recommending further studies or mitigation measures where necessary.

The site encompasses an area of 50 hectares, characterised by a single habitat type classified under Fossitt (2000) as FL3 — Mesotrophic lakes. This habitat is in good condition and holds a county-level evaluation. No specific threats or listed species have been detailed, although the presence of mesotrophic lakes suggests potential ecological interest.

The scope of this ecological assessment includes a planned walkover survey scheduled for 21st February 2026, which will provide an initial understanding of the site's ecological features. Desk research has identified the presence of the Light-bellied Brent Goose, a protected species, within the vicinity, highlighting the need for careful consideration of avian interests.

The assessment is conducted in accordance with relevant Irish and EU legislation, including the Wildlife Acts 1976-2021, which protect native flora and fauna, and the EU Habitats and Birds Directives, which aim to safeguard biodiversity across Europe. The European Communities (Birds and Natural Habitats) Regulations 2011 further implement these directives in Ireland, ensuring the protection of designated sites and species.

The report is structured as follows:

- **Introduction:** Provides the project background, site location, and scope of the ecological assessment.
- **Methodology:** Details the methods employed during the field surveys and desk research.
- **Results:** Presents the findings from the field survey and desk research, including habitat descriptions and species observations.
- **Discussion:** Interprets the results in the context of relevant legislation and potential ecological impacts.
- **Recommendations:** Offers guidance on further surveys, mitigation measures, and any necessary actions to comply with ecological regulations.
- **Conclusion:** Summarises the key findings and outlines the next steps in the ecological assessment process.

This PEA serves as an initial step in understanding the ecological context of the proposed development site. Given the limited data available, particularly regarding species observations and potential impacts on designated sites such as the nearby Castlemaine Harbour SAC, further detailed surveys and assessments may be required to ensure compliance with conservation objectives and legislative requirements.

2. Methodology

Desk Study

The desk study involved a comprehensive review of existing ecological data from several authoritative sources to inform the Preliminary Ecological Appraisal. The following data sources were consulted:

- **National Parks and Wildlife Service (NPWS):**No specific records were retrieved from this source for the site.
- **Global Biodiversity Information Facility (GBIF):**No specific records were retrieved from this source for the site.
- **National Biodiversity Data Centre (NBDC):**Two records of the Light-bellied Brent Goose, a protected species, were identified.
- **Environmental Protection Agency (EPA):**No specific records were retrieved from this source for the site.
- **Catchments.ie:**No specific records were retrieved from this source for the site.

The desk study aimed to identify any protected species, habitats, and designated sites within proximity to the project area, with particular attention to potential impacts on the Castlemaine Harbour SAC.

Field Survey

A walkover survey was planned for the site on 21st February 2026. The survey was designed to assess the ecological value of the site and identify any potential constraints to development. The survey was conducted by a qualified ecologist following standard guidelines as set out by the Chartered Institute of Ecology and Environmental Management (CIEEM) and the National Roads Authority (NRA).

- **Survey Date:**21st February 2026
- **Survey Method:**Walkover survey
- **Weather Conditions:**Not specified in the provided data
- **Surveyor Effort:**Specific start and end times were not provided

The survey aimed to verify the presence of habitats identified in the desk study and to assess the condition of these habitats on-site.

Habitat Mapping

Habitat mapping was conducted in accordance with the Fossitt (2000) classification system. The site was identified as containing one habitat type:

- **FL3 — Mesotrophic lakes:**Covering an area of 50.00 hectares, the habitat is in good condition and evaluated at the county level. It is associated with EU Annex I habitat code 5555.

The mapping process involved a detailed assessment of the habitat's physical and biological characteristics, with particular attention to any potential threats or listed species.

Limitations

Several limitations were identified during the assessment process:

- **Seasonal Constraints:**The survey was conducted in February, a time when certain species may not be present or easily detectable, potentially affecting the comprehensiveness of species observations.
- **Access Restrictions:**No specific access issues were reported, but general access limitations can impact the ability to conduct thorough surveys.
- **Weather Conditions:**The absence of detailed weather data may affect the interpretation of survey results, particularly for species sensitive to weather conditions.

- **Data Gaps:** The lack of specific species observations and detailed weather conditions during the field survey highlights a need for further targeted surveys to ensure a comprehensive ecological assessment.

Further surveys, particularly during different seasons, are recommended to address these limitations and provide a more complete understanding of the site's ecological value.

3. Results

3.1 Designated Sites

Summary Table of Designated Sites

Site Name	Designation	Distance from Site Boundary
Castlemaine Harbour SAC	SAC	Within 2km

Castlemaine Harbour SAC (000343)

Castlemaine Harbour SAC is located within 2km of the project site. This Special Area of Conservation is of significant ecological importance due to its diverse range of habitats and species. The qualifying interest habitats include:

- **[1130] Estuaries**(Condition: Inadequate)
- **[1140] Mudflats and sandflats not covered by seawater at low tide**(Condition: Inadequate)
- **[1210] Annual vegetation of drift lines**(Condition: Inadequate)
- **[1220] Perennial vegetation of stony banks**(Condition: Inadequate)
- **[1230] Vegetated sea cliffs of the Atlantic and Baltic Coasts**(Condition: Favourable)
- **[1310] Salicornia and other annuals colonizing mud and sand**(Condition: Inadequate)
- **[1330] Atlantic salt meadows (*Glauco-Puccinellietalia maritima*)**(Condition: Inadequate)
- **[1410] Mediterranean salt meadows (*Juncetalia maritimi*)**(Condition: Inadequate)

The site faces numerous pressures and threats, including pollution, development, aquaculture, recreation, and climate change. These factors contribute to the inadequate condition of several habitats. The conservation of Castlemaine Harbour is critical for maintaining the ecological integrity of the Natura 2000 network, as it supports Annex II species such as the Atlantic salmon and otter.

Potential Connectivity

The proximity of the project site to Castlemaine Harbour SAC suggests potential ecological connectivity, particularly for mobile species such as birds and aquatic organisms. The presence of mesotrophic lakes (FL3) on the project site could serve as a habitat or stopover point for species associated with the SAC, such as the Light-bellied Brent Goose. Therefore, an Appropriate Assessment (AA) Screening is recommended to evaluate potential impacts on the SAC.

3.2 Habitats

Summary of Habitat Types

- **FL3 — Mesotrophic Lakes**
 - **Area:**50.00 hectares
 - **Condition:**Good
 - **EU Annex I:**5555
 - **Evaluation:**County level

Description and Features

The entire project site comprises a mesotrophic lake habitat (FL3), covering an area of 50 hectares. This habitat is in good condition and is evaluated at the county level for its ecological significance. Mesotrophic lakes are characterised by moderate nutrient levels, supporting a diverse range of aquatic flora and fauna. The habitat is potentially connected to the Castlemaine Harbour SAC, providing ecological corridors for species movement.

Threats

No specific threats were identified during the surveys; however, potential pressures such as eutrophication and invasive species colonisation should be monitored.

3.3 Flora & Invasive Species

Flora Observations

No specific plant species were recorded during the field surveys. The mesotrophic lake habitat is expected to support a variety of aquatic and semi-aquatic plant species typical of such environments.

Invasive Species

No invasive species were observed during the desk study or field survey. However, the mesotrophic lake habitat has the potential for colonisation by invasive species listed in the Third Schedule of the European Communities (Birds and Natural Habitats) Regulations 2011. Regular monitoring is recommended to prevent potential ecological impacts.

3.4 Fauna

Fauna Observations

No direct fauna observations were recorded during the field survey. However, desk research identified the presence of the Light-bellied Brent Goose, a protected species under the Wildlife Acts and the EU Birds Directive. This species is known to utilise habitats similar to those found on the project site.

Protected Species

- **Light-bellied Brent Goose**
 - **Protection:**Wildlife Acts, EU Birds Directive
 - **Observations:**Recorded in desk study

Ecological Importance

The presence of the Light-bellied Brent Goose highlights the potential ecological importance of the site for avian species, particularly those associated with wetland and aquatic habitats. The mesotrophic lake may serve as a feeding or resting area for migratory birds.

Recommendations

Given the proximity to Castlemaine Harbour SAC and the potential presence of protected species, further surveys are recommended to confirm species presence and assess potential impacts. An AA Screening is advised to ensure compliance with relevant conservation legislation and directives.

Data Gaps

- Lack of detailed species-specific surveys for flora and fauna
- Absence of invasive species monitoring data

Further ecological surveys are necessary to address these data gaps and inform appropriate mitigation measures.

4. Ecological Constraints

Ecological Constraints

The following section outlines the ecological constraints identified for the date test project located in Glenbeigh, County Kerry. The assessment is based on a preliminary walkover survey and desk-based research. The ecological value of the site and its surroundings has been evaluated using criteria such as naturalness, rarity, size/extent, diversity, fragility, and typicalness. The Zone of Influence (ZOI) concept has been applied to determine the potential impacts on ecological receptors.

Constraints Table

Ecological...	Import...	Potential Impact	Recommended Action
Mesotrophi...	County	Habitat degradation due to...	Avoidance of direct impacts; buffer zones; timing restri...
Castlemain...	Internat...	Indirect impacts from pollu...	Implement mitigation measures to minimise pollution; ...
Light-bellie...	Internat...	Disturbance from increase...	Avoidance of disturbance during key periods; potential...

Assessment of Ecological Value

- **Mesotrophic Lakes (FL3):**The site comprises 50 hectares of mesotrophic lakes, classified under Fossitt (2000) as FL3. These lakes are in good condition and hold county-level importance due to their size and ecological characteristics. The lakes contribute to local biodiversity and provide habitat for various aquatic species. The naturalness and typicalness of this habitat are significant, though no specific rare species have been recorded.
- **Castlemaine Harbour SAC:**This designated site holds international importance, being part of the Natura 2000 network. It supports a range of Annex I habitats and Annex II species, including estuaries and salt meadows. The SAC is subject to pressures such as pollution and development, which could be exacerbated by project activities. The site's fragility and diversity are high, necessitating careful

management to avoid adverse impacts.

- **Light-bellied Brent Goose:** Although no direct observations were recorded during the survey, desk research indicates the presence of this protected species in the area. The species is of international importance, and potential disturbance from the project could impact its behaviour and habitat use.

Recommended Actions

- **Avoidance and Minimisation:** Direct impacts on the mesotrophic lakes should be avoided by establishing buffer zones and implementing timing restrictions to prevent disturbance during sensitive periods. Construction activities should be planned to avoid the breeding season and other critical times for local wildlife.
- **Remediation and Compensation:** If impacts on the Light-bellied Brent Goose cannot be avoided, compensatory measures such as habitat enhancement or creation may be necessary. Monitoring should be conducted to assess the effectiveness of mitigation measures.
- **Pollution Control:** To protect the Castlemaine Harbour SAC, strict pollution control measures should be implemented. This includes managing runoff and ensuring compliance with the Water Framework Directive to maintain water quality.

Timing Constraints and Seasonal Restrictions

- Construction activities should be scheduled outside of sensitive periods for local wildlife, particularly during the breeding season for birds and other fauna. The specific timing of these restrictions will depend on further detailed surveys and consultations with relevant stakeholders.

Data Gaps and Further Work

- The absence of species observations during the initial survey highlights a data gap that requires further investigation. Additional surveys, particularly during different seasons, are recommended to obtain a comprehensive understanding of the site's ecological value and to inform mitigation strategies effectively.
- Detailed hydrological assessments may be necessary to evaluate potential impacts on the Castlemaine Harbour SAC and to develop appropriate mitigation measures.

In conclusion, the ecological constraints identified necessitate careful planning and implementation of mitigation measures following the mitigation hierarchy to ensure compliance with relevant Irish and EU legislation. Further surveys and assessments are essential to address data gaps and refine the project's ecological management plan.

5. Discussion & Conclusions

Synthesis

The Preliminary Ecological Appraisal for the date test project (Site Code: DTP-2026-763) located in Glenbeigh, County Kerry, has identified key ecological features within the 50-hectare site. The primary habitat type present is classified as FL3 — Mesotrophic lakes, which is in good condition and holds county-level ecological value. The habitat is not listed under EU Annex I, but its condition suggests it plays a significant role in local biodiversity. The desk study revealed the presence of the Light-bellied Brent Goose, a protected species, indicating the potential importance of the site for avian biodiversity.

Potential Impacts

The proposed development could potentially impact the mesotrophic lake habitat through direct habitat loss, alteration of hydrological regimes, and increased nutrient input leading to eutrophication. The presence of the Light-bellied Brent Goose suggests that the site may serve as a feeding or resting area for this species, and any disturbance during critical periods could have adverse effects. Additionally, potential indirect impacts on the nearby Castlemaine Harbour SAC, particularly through water pollution and sedimentation, should be considered given its proximity and ecological significance.

Cumulative Effects

Cumulative impacts must be assessed in conjunction with other developments in the area. The Castlemaine Harbour SAC is already under pressure from pollution, development, and recreational activities. Any additional stressors from the proposed project could exacerbate these pressures, particularly concerning water quality and habitat degradation. The potential in-combination effects with other local projects could lead to significant ecological consequences, necessitating a comprehensive cumulative impact assessment.

Precautionary Principle

Given the uncertainties surrounding the potential impacts on the mesotrophic lake habitat and the Light-bellied Brent Goose, the precautionary principle should be applied. This approach is particularly relevant where data gaps exist, such as the lack of specific species observations during field surveys. The precautionary principle mandates that in the face of uncertainty, measures should be taken to avoid or minimise harm to the environment.

AA Screening

The proximity of the project site to Castlemaine Harbour SAC triggers the need for Appropriate Assessment (AA) Screening. The potential for significant effects on the SAC's qualifying interests, particularly through hydrological connectivity and potential pollution pathways, warrants a detailed screening process. This will determine whether a full Appropriate Assessment is required under the EU Habitats Directive.

Further Surveys

To address data gaps and inform impact assessments, further ecological surveys are recommended:

- **Avian Surveys:** Targeting the Light-bellied Brent Goose and other waterfowl, conducted during winter months (November to February) to capture peak usage periods.
- **Aquatic Surveys:** Assessing water quality and aquatic biodiversity within the mesotrophic lake, optimally conducted in spring and summer (April to August).
- **Habitat Surveys:** Detailed botanical surveys to confirm the condition and extent of the mesotrophic lake habitat, best undertaken in late spring to early summer (May to July).

Enhancement Opportunities

To achieve biodiversity net gain, the following enhancement measures are recommended:

- **Habitat Creation:** Establish buffer zones with native vegetation around the lake to enhance habitat connectivity and provide additional resources for wildlife.
- **Water Quality Improvement:** Implement sustainable drainage systems (SuDS) to manage surface water runoff and reduce nutrient input into the lake.

- **Monitoring Programme:** Develop a long-term ecological monitoring programme to track changes in habitat condition and species presence, ensuring adaptive management strategies can be applied as necessary.

In conclusion, while the date test project presents potential ecological impacts, careful planning and implementation of mitigation and enhancement measures can minimise adverse effects and contribute positively to local biodiversity. Further surveys and AA Screening will be crucial in refining these strategies and ensuring compliance with relevant Irish and EU environmental legislation.

6. Appendices

Appendix A: Site Location Map

- This appendix includes a detailed map illustrating the geographical location of the project site within Glenbeigh, County Kerry. The map provides the grid reference A 535 750 for precise identification.

Appendix B: Habitat Map

- The habitat map outlines the extent of the mesotrophic lake (FL3) covering 50 hectares. The map references FOSSITT (2000) habitat classification codes and indicates the condition of the habitat as good.

Appendix C: Species List

- This appendix contains a list of species recorded during desk research, specifically the Light-bellied Brent Goose, which is protected. No species were observed during field surveys; hence, observation counts are not applicable.

Appendix D: Site Photographs

- A collection of photographs taken during the planned walkover survey on 21st February 2026. These images provide visual documentation of the site conditions and habitat types present.

Appendix E: Survey Datasheets

- Datasheets from the field survey conducted on 21st February 2026 are included here. These datasheets detail the methodologies used and any relevant observations made during the survey.

Caveats:

- The absence of species observations during the field survey highlights the need for further targeted surveys to comprehensively assess the site's biodiversity.

Appendix A: Habitat Map

[Content to be inserted]

Appendix B: Species List

Name	AI Sum...	Protection Status
Branta bernicla hrota (O...		Protected Species: Wildlife Acts Threatened Species: Birds of Conservati...

Appendix C: Site Photographs

[Content to be inserted]